

A counterexample to Kaplansky's quasitrace conjecture and failure of tensor-product stable finiteness

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September 23, 2026

Abstract

We refute Kaplansky's quasitrace conjecture by constructing a separable unital complex C^* -algebra that admits normalized 2-quasitraces but has no tracial state. As a consequence, we show that the minimal tensor product of two unital simple stably finite complex C^* -algebras can be properly infinite, even when one factor is the reduced free-group algebra $C_r^*(\mathbb{F}_2)$.

1 Introduction

A unital C^* -algebra is *stably finite* when every matrix algebra over it has no nonunitary isometry, and is *properly infinite* when it contains two isometries with orthogonal ranges. Milhøj and Rørdam asked whether the minimal tensor product of two unital simple stably finite C^* -algebras must again be stably finite [8, Question 2.5]. They proved that this question is equivalent to the quasitrace problem and that a negative answer to that problem would yield a properly infinite tensor product with $C_r^*(\mathbb{F}_\infty)$ [8, Section 2 and Theorem 3.1]. We construct the required quasitrace counterexample. The resulting tensor product can be properly infinite with one factor equal to the reduced C^* -algebra $C_r^*(\mathbb{F}_2)$ of the free group on two generators.

A quasitrace retains positivity and the identity $\tau(x^*x) = \tau(xx^*)$, but is required to be linear only on commutative subalgebras, together with compatibility with real and imaginary parts. A 2-quasitrace must also extend to the two-by-two matrix algebra. Whether this extra condition forces full linearity is the general quasitrace problem. It is equivalent to Kaplansky's question whether every AW^* -factor of type II_1 is a von Neumann algebra; see [8, Section 2] and [4, Conjectures 1.1–1.2 and Theorem 4.11]. Here the term *von Neumann algebra* has its abstract W^* meaning: the C^* -algebra admits a Banach-space predual.

Gow's historical account explains that Kaplansky introduced AW^* -algebras in 1951 to characterize W^* -algebras by intrinsic algebraic conditions, independent of a chosen Hilbert-space representation. The factor types have different outcomes: every type I AW^* -factor is a W^* -algebra, whereas type III counterexamples exist. The type II_1 factor question is the finite case addressed here [4, Introduction].

Quasitraces also connect the projection-theoretic origins of the question to dimension theory for general C^* -algebras. Cuntz developed dimension functions for simple algebras [3], and Blackadar and Handelman established the correspondence between 2-quasitraces and lower semicontinuous dimension functions [1]. Haagerup credits Blackadar and Handelman with proving that a positive answer to Kaplansky's factor question would imply that every 2-quasitrace on a unital C^* -algebra is a trace. He also recalls their realization of such quasitraces as pullbacks of quasitraces on finite AW^* -algebras along unital $*$ -homomorphisms

[6, Introduction and Proposition 3.3]. Blackadar and Rørdam later developed the extension of states on preordered semigroups that underlies the general existence criterion [2].

The exact case was settled by Haagerup: every 2-quasitrace on a unital exact C^* -algebra is a trace [6, Theorem 5.11]. A first draft was completed in 1991 and circulated thereafter; the paper was published in 2014. Without exactness, the existence of quasitraces is governed by a different condition. The Cuntz–Blackadar–Handelman existence theorem provides a normalized 2-quasitrace whenever no matrix algebra over the given unital algebra is properly infinite [8, Theorem 2.3]. Thus a unital algebra with that property but without tracial states would answer the general question negatively.

We construct such an algebra and obtain a fixed positive pair witnessing nonadditivity. Throughout, a normalized 2-quasitrace has value one on the unit of its domain; its corner-preserving extension to M_2 has value two on the matrix identity. The definitions are recalled in Section 2.

Theorem 1.1. *There exist a separable unital complex C^* -algebra A , admitting a normalized 2-quasitrace, and positive contractions $a, b \in A$ such that every normalized 2-quasitrace τ on A satisfies*

$$\tau(a + b) - \tau(a) - \tau(b) \geq \frac{1}{144}.$$

Theorem 1.1 refutes the conjecture that every normalized 2-quasitrace on every unital complex C^* -algebra is a tracial state. By the equivalence recalled above, it also gives a negative answer to Kaplansky’s II_1 AW^* -factor question. The constructed algebra is necessarily nonexact, by Haagerup’s result.

The matrix-extension requirement is essential to this conclusion. Haagerup reports Kirchberg’s 2006 unpublished example of a 1-quasitrace that is not a 2-quasitrace, and hence is not a trace [6, Introduction]. This example lacks the matrix extension required here. Gow’s recent work places the established factor–quasitrace equivalence alongside further equivalent formulations involving maximal abelian subalgebras and bidual retractions [4, Theorem 4.11].

The tensor-product conclusion and two related consequences can be stated together. All tensor products below are minimal C^* -tensor products.

Corollary 1.2 (Tensor products and stable finiteness).

- (i) *There are two unital simple stably finite complex C^* -algebras whose minimal tensor product is properly infinite. One factor can be taken to be $C_r^*(\mathbb{F}_2)$.*
- (ii) *There is a separable unital complex C^* -algebra B admitting a normalized 2-quasitrace such that $B \otimes_{\min} C_r^*(\mathbb{F}_2)$ is properly infinite and admits no normalized 2-quasitrace.*
- (iii) *There is a separable unital stably finite complex C^* -algebra with no tracial state.*

The three assertions use related but different algebras. The simple tensor-product example comes from a non-von-Neumann AW^* -factor of type II_1 , with no separability assertion for that factor. The quasitrace-loss example uses the explicit separable algebra B constructed below; we prove that no matrix algebra over B is properly infinite, not that B is stably finite. A finitely generated subalgebra of the AW^* -factor gives the separable stably finite traceless example, without a simplicity assertion. The short deductions from established trace-obstruction and tensor-product results are given in Section 6.

The construction

We first build a unital algebra $B = C^*(1, x_1, \dots, x_{24})$ such that

$$\sum_{i=1}^{24} x_i^* x_i = 1, \quad \sum_{i=1}^{24} x_i x_i^* \leq \frac{2}{3} 1, \quad (1.1)$$

while no $M_m(B)$, $m \geq 1$, contains two isometries with orthogonal ranges. A tracial state would make the two sums in (1.1) have the same value, so B has none. These relations instantiate Haagerup's characterization of the absence of tracial states by a finite tuple with column sum one and a strict row bound [6, Lemma 2.1]. Milhøj and Rørdam proposed separating a uniformly bounded length of such a tuple from increasingly delayed matrix proper infiniteness [8, Remark 4.7]. The construction below uses a bounded product and a generated subalgebra, with a separate test for every matrix size and polynomial degree. The fixed number of generators preserves the trace obstruction while all finite polynomial candidates are excluded.

The generators are shifts between finite-rank vector bundles. For each pair of positive integers (m, R) , we choose a compact base $Y_{m,R}$ and bundles $E_j \rightarrow Y_{m,R}$ for all $j \geq 0$, with E_0 trivial. Only levels $0, \dots, R$ enter the finite topological test. A polynomial matrix $P \in M_{m,2m}(B)$ whose entries contain only words of length at most R restricts to a bundle map

$$P_0 : E_0^{\oplus 2m} \longrightarrow \left(\bigoplus_{j=0}^R E_j \right)^{\oplus m}.$$

The target's Chern class forbids the trivial image bundle that would result from $P^*P \geq \frac{1}{2}I$. Write $S_i^{m,R}(y)$ for the corresponding forward shift on the Hilbert direct sum $\bigoplus_{j \geq 0} E_j(y)$. Uniform norm bounds allow the simultaneous definitions

$$x_i = (S_i^{m,R}(y))_{m,R \geq 1, y \in Y_{m,R}}, \quad B = C^*(1, x_1, \dots, x_{24}),$$

in one bounded operator product. The number of generators and the row bound in (1.1) are uniform, while the bundle ranks and base dimensions may depend on (m, R) . A polynomial matrix approximating a candidate pair of isometries has the same coefficients at every coordinate. Its finite degree selects one of the tests already present in the product, which obstructs that approximation.

The topological obstruction follows the use of characteristic classes in the constructions of Villadsen [13, Section 3] and Rørdam [10, Proposition 4.5]. Here projective-space factors retain a nonzero Chern class after a controlled trivial summand is allowed. The parameters are selected anew for each finite test.

Two ingredients keep the expansion and the topological obstruction compatible. First, a family of bundle maps whose collective image expands every source subspace by a factor of two can be normalized continuously to have column sum one and row sum at most $2/3$. The determinant-potential approach is related to operator scaling [5, Section 4] and to geodesically convex matrix optimization [12, Corollary 2.11]. We prove strict convexity and existence of a unique minimizer for the present rectangular maps; uniqueness makes the normalization compatible with changes of bundle frame. Second, the bundle tests separate small and large source subspaces: sixteen maps into reduced copies of old line bundles handle the small ones, while eight generic maps into a new line-bundle block handle the large ones. This keeps the generator count fixed as the matrix size and word-length bound increase.

Finally, set $A = M_{48}(B)$ and apply Theorem 2.1 directly to A . A finite telescoping construction, using only linearity on commuting elements, gives the pair in Theorem 1.1. This last step works for every normalized 1-quasitrace on A , hence for every normalized 2-quasitrace there, and requires no formula for a chosen quasitrace.

Organization

Section 2 records the operator-algebra conventions and the existence theorem. Section 3 proves the incidence and scaling lemmas. The finite bundle tests are assembled in Section 4. Section 5 constructs B , excludes proper infiniteness in every matrix size, exhibits the positive pair, and completes the proof of Theorem 1.1. Section 6 proves Corollary 1.2.

2 Quasitraces and proper infiniteness

All C^* -algebras in this paper are complex, and all matrix algebras carry their usual C^* -norms. For a unital C^* -algebra C , a 1-*quasitrace* is a function $\tau : C \rightarrow \mathbb{C}$ satisfying

- (i) $\tau(x^*x) = \tau(xx^*) \geq 0$ for every $x \in C$;
- (ii) $\tau(h + ik) = \tau(h) + i\tau(k)$ for self-adjoint $h, k \in C$;
- (iii) the restriction of τ to every abelian C^* -subalgebra of C is complex linear.

It is a 2-*quasitrace* if a 1-quasitrace on $M_2(C)$ restricts to τ on the upper-left corner. It is *normalized* when $\tau(1_C) = 1$. The extension then has value two at $1_{M_2(C)}$: the two diagonal corner units have the same value by (i) and their values add by (iii).

A unital C^* -algebra C is *properly infinite* when it contains isometries s_1, s_2 with orthogonal ranges. Equivalently,

$$s_i^* s_j = \delta_{ij} 1_C, \quad i, j \in \{1, 2\}.$$

It is *stably properly infinite* if $M_m(C)$ is properly infinite for some positive integer m . Absence of stable proper infiniteness is the condition needed below; no assertion about finiteness of every projection is required.

Theorem 2.1 (Quasitrace existence). *A unital C^* -algebra C admits a normalized 2-quasitrace if and only if $M_m(C)$ is not properly infinite for every positive integer m .*

We use Theorem 2.1 in the formulation of [8, Theorem 2.3], with the normalization and matrix-extension conventions fixed at the start of its Section 2. Cuntz's work on dimension functions [3] precedes the general criterion. The correspondence between 2-quasitraces and lower semicontinuous dimension functions is due to Blackadar–Handelman [1, Theorems II.2.2 and II.3.1]; the extension of semigroup states underlying this formulation is discussed in [2]. In particular, Theorem 2.1 has no exactness, simplicity, or representation assumption.

For finite-rank complex vector bundles, Hermitian metrics are fixed. Adjoints, positive endomorphisms, and their square roots are taken fiberwise. A positive lower bound on the adjoint square of a continuous bundle map makes it injective with a continuous image subbundle: if W^*W is invertible, then

$$Q = W(W^*W)^{-1/2}, \quad Q^*Q = I,$$

has the same image, and QQ^* is its continuous orthogonal projection. We will use this observation when passing from polynomial operators to the Chern-class obstruction.

3 Two bundle-map lemmas

We first construct maps whose collective images expand subspaces. We then normalize such a family by a fiberwise minimization. Uniqueness of the minimizer will make the normalization continuous on the base, even when the bundles are nontrivial.

Lemma 3.1 (Avoiding small collective images). *Let E, F be smooth Hermitian complex vector bundles of ranks d, r over a compact smooth manifold Y of real dimension $2h$, and let $q \geq 1$. Let $\mathcal{A}$ be a finite set of integer pairs (s, t) with $1 \leq s \leq d$ and $0 \leq t \leq r$ such that*

$$(qs - t)(r - t) - s(d - s) > h \quad ((s, t) \in \mathcal{A}). \quad (3.1)$$

There are smooth bundle maps $T_1, \dots, T_q : E \rightarrow F$ such that, for every $y \in Y$, every $(s, t) \in \mathcal{A}$, and every complex s -plane $S \subset E_y$,

$$\dim_{\mathbb{C}} \sum_{i=1}^q T_i(y)S > t.$$

Here and below the sum denotes linear span. The same statement holds over a point, with $h = 0$.

Proof. Choose a finite cover trivializing $\text{Hom}(E, F)$ and a smooth partition of unity subordinate to it, with each support contained in its trivializing open set. Multiply every local frame section by the corresponding partition function and extend by zero. Their complex linear span is a finite-dimensional space V of smooth bundle maps whose evaluation map

$$V \longrightarrow \text{Hom}(E_y, F_y)$$

is surjective for every y : at least one partition function is nonzero there, and its multiplied frame still spans the fiber. Thus the space $\mathcal{V} = V^q$, of some complex dimension M , parametrizes tuples with surjective evaluation onto $\text{Hom}(E_y, F_y)^q$.

Fix $(s, t) \in \mathcal{A}$. The Grassmann bundle of triples (y, S, U) , where $S \subset E_y$ is an s -plane and $U \subset F_y$ is a t -plane, has real dimension

$$2h + 2s(d - s) + 2t(r - t).$$

Over this bundle, evaluation followed by restriction and quotient gives a smooth, fiberwise surjective complex-linear map from the trivial bundle with fiber $\mathcal{V}$ to the bundle with fiber $\text{Hom}(S, F_y/U)^q$. Surjectivity follows because any linear map $S \rightarrow F_y/U$ extends to a map $E_y \rightarrow F_y$, and evaluation from V is surjective. The target has complex rank $qs(r - t)$, so the kernels form a smooth vector bundle whose total space $\mathcal{I}_{s,t}$ has real dimension

$$2M + 2h + 2s(d - s) + 2t(r - t) - 2qs(r - t) < 2M,$$

where the inequality is precisely (3.1).

The image of the smooth projection $\mathcal{I}_{s,t} \rightarrow \mathcal{V}$ consists exactly of tuples having some s -plane whose collective image lies in a t -plane. A smooth image of a manifold of smaller dimension is a Lebesgue null set, as in the lower-dimensional case of Sard's theorem [11]. Indeed, on a compact coordinate patch the projection is Lipschitz. Subdivision into cubes of side δ covers its image by $O(\delta^{-a})$ balls of radius $O(\delta)$, where $a < 2M$ is the domain dimension; their total $2M$ -dimensional volume tends to zero. A countable exhaustion by such patches handles the possibly noncompact total space. The finite union of these null sets over $\mathcal{A}$ cannot exhaust $\mathcal{V}$. A tuple outside that union has the required properties. $\square$

The next normalization uses a determinant potential. Gurvits's operator-scaling theory uses determinant capacity to relate rank nondecrease to approximate doubly stochastic scaling [5, Section 4]. Related affine log-determinant functions are studied by Sra and Hosseini [12, Corollary 2.11]. We prove the version needed here, including the strict convexity and continuity that allow normalization over nontrivial bundles.

Lemma 3.2 (Continuous normalization). *Let E, F be Hermitian complex vector bundles of positive ranks over a compact manifold Y . Suppose that continuous bundle maps $T_1, \dots, T_n : E \rightarrow F$ satisfy*

$$\dim_{\mathbb{C}} \sum_{i=1}^n T_i(y)S \geq 2 \dim_{\mathbb{C}} S \quad (y \in Y, S \subset E_y). \quad (3.2)$$

Then there are continuous bundle maps $X_1, \dots, X_n : E \rightarrow F$ with

$$\sum_{i=1}^n X_i^* X_i = I_E, \quad \sum_{i=1}^n X_i X_i^* \leq \frac{2}{3} I_F. \quad (3.3)$$

Proof. Write $d = \text{rank } E$ and $r = \text{rank } F$. In each fiber define

$$D_y(P) = \sum_{i=1}^n T_i(y) P T_i(y)^*, \quad g_y(P) = \log \det(I_{F_y} + D_y(P)) - \frac{3}{2} \log \det P$$

for positive definite $P \in \text{End}(E_y)$. Both determinants are intrinsic. We will prove that g_y has a unique minimizer depending continuously on y . The expansion factor 2 and the coefficient $3/2$ leave a strict margin that prevents this minimizer from escaping to infinity.

Uniform compact sublevels. Order all eigenvalues decreasingly. For an orthogonal rank- s projection p on E_y ,

$$\text{ran } D_y(p) = \sum_i T_i(y) \text{ran } p.$$

Indeed, both sides have orthogonal complement $\bigcap_i \ker(p T_i(y)^*)$. By (3.2), the $2s$ -th eigenvalue of $D_y(p)$ is positive; in particular $r \geq 2d$. Eigenvalues are continuous, and the Grassmann bundle of such projections is compact. Taking a minimum there and then over $1 \leq s \leq d$ gives $\epsilon > 0$ such that

$$\mu_{2s}(D_y(p)) \geq \epsilon$$

for every y, s, p under consideration.

If P has eigenvalues $\lambda_1 \geq \dots \geq \lambda_d > 0$, choose any projection p_s onto s top eigenvectors. Then $P \geq \lambda_s p_s$, so positivity of D_y and eigenvalue monotonicity give

$$\mu_{2s}(D_y(P)) \geq \epsilon \lambda_s.$$

Pairing the target eigenvalues with indices $2s - 1, 2s$, and discarding any remaining nonnegative terms, yields

$$\log \det(I + D_y(P)) \geq 2 \sum_{s=1}^d \log(1 + \epsilon \lambda_s). \quad (3.4)$$

No continuous choice of the projections p_s is needed.

Set $f(t) = 2 \log(1 + \epsilon t) - \frac{3}{2} \log t$. This function is bounded below and tends to $+\infty$ at both endpoints of $(0, \infty)$. Let $b = \min f$ and $C = \max_{y \in Y} g_y(I_{E_y})$. Equation (3.4) shows that, whenever $g_y(P) \leq C$,

$$f(\lambda_j) \leq C - (d - 1)b \quad (1 \leq j \leq d).$$

Thus all eigenvalues of every such P lie in a common compact interval $[a, A] \subset (0, \infty)$. Outside the compact set $aI_{E_y} \leq P \leq AI_{E_y}$, we have $g_y(P) > C \geq g_y(I_{E_y})$. Minimizing on this set therefore gives a global minimizer in each fiber, lying in the interior of the positive definite cone.

Strict convexity along positive definite geodesics. Fix $P_0 > 0$ and a nonzero Hermitian H in one source fiber, and put

$$P(u) = P_0^{1/2} \exp(uH) P_0^{1/2}.$$

Choose an orthonormal eigenbasis e_j for H , with eigenvalues h_j , and set $v_{ij} = T_i P_0^{1/2} e_j$. Then

$$D(P(u)) = \sum_{i,j} e^{uh_j} v_{ij} v_{ij}^*.$$

The Gram determinant expansion of $\det(I + CC^*)$, applied to the columns $e^{uh_j/2} v_{ij}$, expresses $\det(I + D(P(u)))$ as a finite sum $\sum_{\alpha} c_{\alpha} e^{u\alpha}$ with $c_{\alpha} \geq 0$. The empty subset contributes 1, and singleton subsets contribute $\|v_{ij}\|^2 e^{uh_j}$. For every j , the total singleton coefficient at h_j is positive: the vector $P_0^{1/2} e_j$ is nonzero and (3.2) makes the common kernel of the maps T_i trivial. Since some $h_j \neq 0$, the expansion contains positive terms at two distinct exponents, 0 and h_j .

The second derivative of the logarithm of a positive exponential sum is the variance of its exponents under weights proportional to $c_{\alpha} e^{u\alpha}$. It is therefore strictly positive here. On the other hand,

$$\log \det P(u) = \log \det P_0 + u \operatorname{tr} H$$

is affine. Hence $g(P(u))$ is strictly convex. Any two distinct positive definite matrices P_0, P_1 are joined by such a nonconstant curve, by taking $H = \log(P_0^{-1/2} P_1 P_0^{-1/2})$. The minimizer is unique.

Continuity and gluing. In local unitary frames, $(y, P) \mapsto g_y(P)$ is continuous and the minimizers lie in the common compact matrix set $aI \leq P \leq AI$. If $y_k \rightarrow y$, every subsequential limit of their minimizers minimizes g_y : pass to the limit in the minimizing inequality against any fixed positive definite comparison matrix. Uniqueness forces every such limit to be the minimizer at y , proving continuity. On overlaps of unitary frames, P and $I_{F_y} + D_y(P)$ change by unitary conjugation, preserving both determinants. Uniqueness therefore makes the local minimizers glue to a continuous positive definite endomorphism P of E .

The normalization. Put $Q = I_F + D(P)$. Differentiation at the minimizer in an arbitrary Hermitian source direction K gives

$$0 = \operatorname{tr} \left[\left(\sum_i T_i^* Q^{-1} T_i - \frac{3}{2} P^{-1} \right) K \right].$$

The trace pairing on Hermitian matrices is nondegenerate, so $\sum_i T_i^* Q^{-1} T_i = \frac{3}{2} P^{-1}$. Define

$$X_i = \sqrt{\frac{2}{3}} Q^{-1/2} T_i P^{1/2}.$$

Continuous functional calculus makes these continuous bundle maps, and the order of the factors gives

$$\begin{aligned}\sum_i X_i^* X_i &= \frac{2}{3} P^{1/2} \left(\sum_i T_i^* Q^{-1} T_i \right) P^{1/2} = I_E, \\ \sum_i X_i X_i^* &= \frac{2}{3} Q^{-1/2} D(P) Q^{-1/2} = \frac{2}{3} (I_F - Q^{-1}) \leq \frac{2}{3} I_F.\end{aligned}$$

These are (3.3). □

4 Bundle tests of bounded degree

We now construct the finite topological tests used to exclude polynomial witnesses of proper infiniteness. The matrix size m and the degree bound R determine the bundles and their base, while the number of maps remains fixed at 24. The use of characteristic classes to obstruct comparison follows the methods of Villadsen [13, Section 3] and Rørdam [10, Sections 3–4]. Our test retains enough of a Chern class to exclude a specified trivial subbundle, even though the maps between successive levels have the normalized column and row bounds.

Proposition 4.1. *For every pair of integers $m, R \geq 1$, set*

$$d_0 = 64m4^R, \quad d_j = 4^j d_0 \quad (0 \leq j \leq R). \quad (4.1)$$

There are a compact smooth manifold Y , Hermitian complex vector bundles $E_j \rightarrow Y$ for all $j \geq 0$, and continuous bundle maps $X_i^{(j)} : E_j \rightarrow E_{j+1}$, for $1 \leq i \leq 24$ and $j \geq 0$, such that E_0 is trivial of rank d_0 , the rank of E_j is d_j for $0 \leq j \leq R$, and

$$\sum_{i=1}^{24} (X_i^{(j)})^* X_i^{(j)} = I_{E_j}, \quad \sum_{i=1}^{24} X_i^{(j)} (X_i^{(j)})^* \leq \frac{2}{3} I_{E_{j+1}}. \quad (4.2)$$

Moreover,

$$\left(\bigoplus_{j=0}^R E_j \right)^{\oplus m} \quad (4.3)$$

contains no trivial subbundle of rank $2md_0$.

Proof. Fix the matrix size m and degree bound R . The bundles must satisfy two requirements. Their sum through level R must retain a Chern class after allowing the trivial directions coming from E_0 . At the same time, successive bundles must admit a fixed number of maps that expand every source subspace. We reduce each old line-bundle multiplicity by a factor of four at every step, so the total trivial part stays small. Each new projective-space factor is then sized to retain the Chern class contributed by every later occurrence of its line bundle. The initial rank leaves enough room in the reduced blocks to expand small subspaces; a large new block handles the others.

The bundles and their base. For $1 \leq i \leq R$, define the fresh multiplicities

$$b_i = \frac{15}{4} d_{i-1}.$$

Let the multiplicity of the line bundle L_i in E_j be

$$\begin{aligned}a_{0,j} &= d_0 4^{-j} = 64m4^{R-j}, \\ a_{i,j} &= b_i 4^{-(j-i)} = 960m4^{R+2i-j-2} \quad (1 \leq i \leq j \leq R).\end{aligned} \quad (4.4)$$

All these numbers are positive integers. Put

$$K_i = m \sum_{j=i}^R a_{i,j} = 320m^2 4^{2i-2} (4^{R-i+1} - 1), \quad Y = \prod_{i=1}^R \mathbb{CP}^{K_i}. \quad (4.5)$$

Thus every K_i is a positive integer. Take L_0 to be the trivial line bundle on Y , and for $i \geq 1$ let L_i be the pullback of the hyperplane line bundle from the i th factor. Choose Hermitian metrics and define

$$E_j = \bigoplus_{i=0}^j L_i^{\oplus a_{i,j}} \quad (0 \leq j \leq R).$$

In passing from E_j to E_{j+1} , each old multiplicity is divided by four and a fresh block of rank $b_{j+1} = (15/4)d_j$ is added. Consequently the ranks satisfy

$$\text{rank } E_{j+1} = \frac{1}{4}d_j + \frac{15}{4}d_j = 4d_j,$$

which proves the rank assertion by induction. Formula (4.4) also shows that every multiplicity D to which division by four is applied is divisible by four and satisfies $D \geq 256m$.

The retained Chern class. The choice of K_i records exactly the total multiplicity of L_i in the bundle to be tested. Set

$$G = \left(\bigoplus_{j=0}^R E_j \right)^{\oplus m}, \quad h_0 = m \sum_{j=0}^R a_{0,j}, \quad p = \sum_{i=1}^R K_i.$$

The definitions of the multiplicities and K_i give

$$G \cong L_0^{\oplus h_0} \oplus \bigoplus_{i=1}^R L_i^{\oplus K_i}, \quad h_0 = \frac{4}{3}md_0(1 - 4^{-(R+1)}) < 2md_0. \quad (4.6)$$

Let $z_i = c_1(L_i)$. The integral cohomology ring is

$$H^*(Y; \mathbb{Z}) \cong \mathbb{Z}[z_1, \dots, z_R] / (z_1^{K_1+1}, \dots, z_R^{K_R+1}), \quad \deg z_i = 2.$$

In particular $\prod_i z_i^{K_i}$ is nonzero. The Whitney sum formula and the normalization for line bundles give

$$c_p(G) = \prod_{i=1}^R z_i^{K_i} \neq 0;$$

we use the standard Chern-class properties in [7, Theorem 3.2]. If G had a trivial subbundle of rank $k > h_0$, its Hermitian complement would have rank $h_0 + p - k < p$. The Whitney sum formula would identify $c_p(G)$ with the p th Chern class of that complement, which vanishes above its rank. This contradiction, together with (4.6), proves (4.3). The argument applies to any continuous subbundle, without a condition on its position relative to the displayed line summands.

The bases for the expanding maps. It remains to construct the maps between these bundles. For the step $j \rightarrow j+1$, with $0 \leq j < R$, the source and target use only $L_0, \dots, L_{j+1}$ and are pulled back from

$$Y_{j+1} = \prod_{i=1}^{j+1} \mathbb{CP}^{K_i}.$$

The dimensions K_i here are those in (4.5), including all later occurrences of L_i through level R . The complex dimension η_j of this smaller base satisfies

$$\begin{aligned}\eta_j &= \sum_{i=1}^{j+1} K_i \leq \frac{4}{3}m \sum_{i=1}^{j+1} b_i \\ &= \frac{5}{3}md_0(4^{j+1} - 1) < \frac{20}{3}md_j.\end{aligned}\tag{4.7}$$

Here the first inequality uses the geometric series with ratio $1/4$. We construct the maps for this step on Y_{j+1} , and then pull them back to Y . The estimate by a multiple of md_j is what will allow the incidence argument to handle all subspaces of dimension greater than m , even though the full base Y may be much larger.

Sixteen maps for small subspaces. Decompose E_{j+1} as the sum of its replacement blocks and its fresh block. On a source block $L_i^{\oplus D}$, the corresponding replacement block is $L_i^{\oplus(D/4)}$. We seek 16 scalar matrices between the multiplicity spaces and tensor them with the identity of L_i . For $1 \leq s \leq m$, set $r = D/4$ and $t = 2s$. Since $D \geq 64m$,

$$t < r, \quad r - 2s \geq D/8.$$

The incidence expression of Lemma 3.1, with $q = 16$ and base a point, is strictly positive:

$$(16s - 2s)(D/4 - 2s) - s(D - s) = \frac{5}{2}sD - 27s^2 \geq 133ms > 0.\tag{4.8}$$

Applying Lemma 3.1 simultaneously for $s = 1, \dots, m$ gives matrices whose images of every such s -plane span a space of dimension at least $2s$. In fact, avoiding dimension at most $2s$ gives a slightly stronger conclusion. Make these choices for each block and assemble them into 16 block diagonal maps $T_1, \dots, T_{16}$ to the replacement part, with zero component in the fresh block.

We verify that the same 16 maps expand every subspace of the whole source of dimension at most m . Work in one fiber and write the ordered source and replacement blocks as V_k and F_k . Let $V_{\leq k}$ and $F_{\leq k}$ denote their initial sums. Given a subspace S of dimension at most m , set

$$S_k = S \cap V_{\leq k}, \quad W = \sum_{i=1}^{16} T_i S, \quad W_k = W \cap F_{\leq k}.$$

Use the zero spaces before the first flag terms. Projection onto the k th block identifies S_k/S_{k-1} and W_k/W_{k-1} with subspaces of V_k and F_k , respectively. Block diagonality gives $T_i(S_k) \subseteq W_k$ and hence

$$\sum_{i=1}^{16} T_{i,k}(S_k/S_{k-1}) \subseteq W_k/W_{k-1},$$

where $T_{i,k}$ is the map on the k th block. Each source quotient has dimension at most m , so the block estimates yield

$$\dim W = \sum_k \dim(W_k/W_{k-1}) \geq 2 \sum_k \dim(S_k/S_{k-1}) = 2 \dim S.\tag{4.9}$$

This proves the required estimate even when S has correlated components in several blocks.

Eight maps for large subspaces. Let $d = d_j$ and let the target now be the fresh block of rank $r = (15/4)d$. For every integer $m < s \leq d$, take $q = 8$ and $t = 2s$. The rank endpoint is valid because $2s \leq 2d < r$. The incidence expression satisfies

$$\begin{aligned} (8s - 2s)(15d/4 - 2s) - s(d - s) &= \frac{43}{2}sd - 11s^2 \\ &\geq \frac{21}{2}sd > \frac{20}{3}md > \eta_j. \end{aligned} \tag{4.10}$$

The first inequality follows from $s \leq d$. Lemma 3.1 on Y_{j+1} , applied to the finite list $s = m + 1, \dots, d$, therefore supplies eight smooth maps to the fresh block whose image spans have dimension at least $2s$ for all these subspaces. Regard these as maps to E_{j+1} with zero replacement components.

Together, the two lists give 24 maps $T_i : E_j \rightarrow E_{j+1}$ satisfying

$$\dim \sum_{i=1}^{24} T_i S \geq 2 \dim S$$

for every fiber subspace S . The estimate is supplied by the first list when $\dim S \leq m$ and by the second when $\dim S > m$; adding maps cannot decrease an image span. Pull back to Y and apply Lemma 3.2. This gives continuous maps $X_i^{(j)}$ satisfying (4.2) for every $0 \leq j < R$. The scaling may mix the displayed line summands, but each resulting map still has source E_j and target E_{j+1} .

Continuation beyond the test. For $j \geq R$, define $E_{j+1} = E_j \oplus E_j$, and let

$$X_1^{(j)} = 2^{-1/2} \iota_1, \quad X_2^{(j)} = 2^{-1/2} \iota_2, \quad X_i^{(j)} = 0 \quad (3 \leq i \leq 24),$$

where ι_1, ι_2 are the two inclusions. Their column sum is I_{E_j} and their row sum is $\frac{1}{2}I_{E_{j+1}}$, so (4.2) holds for these steps as well. This continuation leaves the retained bundle in (4.3) unchanged and completes the construction. $\square$

5 The algebra and a uniform additivity defect

We now put all the bundle tests into one algebra. The uniform number of maps and their uniform norm bounds allow this construction, while each test obstructs polynomial candidates at its prescribed matrix size and degree.

Theorem 5.1. *There is a separable unital complex C^* -algebra $B = C^*(1, x_1, \dots, x_{24})$ such that*

$$\sum_{i=1}^{24} x_i^* x_i = 1, \quad \sum_{i=1}^{24} x_i x_i^* \leq \frac{2}{3} 1, \tag{5.1}$$

and $M_m(B)$ is not properly infinite for every integer $m \geq 1$.

Proof. For each pair of integers $m, R \geq 1$, choose the data supplied by Proposition 4.1, and denote its base by $Y_{m,R}$. At a point $y \in Y_{m,R}$ form the Hilbert space

$$\mathcal{H}_{m,R,y} = \bigoplus_{j=0}^{\infty} E_j(y).$$

For $1 \leq i \leq 24$, let $S_i^{m,R}(y)$ be the forward shift on this space whose block from $E_j(y)$ to $E_{j+1}(y)$ is $X_i^{(j)}(y)$. The column identity in Proposition 4.1 gives

$$X_i^{(j)*} X_i^{(j)} \leq I_{E_j},$$

so every block, and hence every shift, has norm at most one. These bounds hold uniformly over all coordinates.

Let $\mathcal{D}$ be the bounded C^* -product of the algebras $\mathcal{B}(\mathcal{H}_{m,R,y})$ over all these indices: its elements are the operator families of finite supremum norm. The uniform bound therefore defines a single tuple in $\mathcal{D}$,

$$x_i = (S_i^{m,R}(y))_{m,R \geq 1, y \in Y_{m,R}} \quad (1 \leq i \leq 24).$$

The column sum of the shifts is the identity on every level. Their row sum is zero on level zero and, on level $j \geq 1$, is

$$\sum_{i=1}^{24} X_i^{(j-1)} X_i^{(j-1)*} \leq \frac{2}{3} I_{E_j}.$$

Thus (5.1) holds in $\mathcal{D}$. Take $B = C^*(1, x_1, \dots, x_{24}) \subseteq \mathcal{D}$. This is a nonzero unital algebra, and it is separable: the $*$ -polynomials in these generators with rational complex coefficients form a countable dense subset.

Fix $m \geq 1$ and suppose, towards a contradiction, that $M_m(B)$ contains isometries s_1, s_2 with orthogonal ranges. Their rectangular concatenation satisfies

$$W = [s_1 \ s_2] \in M_{m,2m}(B), \quad W^*W = I_{2m}.$$

By norm density choose $P \in M_{m,2m}(B)$ with $*$ -polynomial entries such that $\|P - W\| < 1/5$. Since $\|W\| = 1$,

$$\|P^*P - I_{2m}\| \leq 2\|P - W\| + \|P - W\|^2 < \frac{1}{2},$$

and consequently

$$P^*P \geq \frac{1}{2} I_{2m}. \quad (5.2)$$

After fixing this approximation, choose $R \geq 1$ at least the length of every word occurring in its entries. We use the coordinate family belonging to the already constructed test (m, R) .

A word of length at most R , applied to level zero, takes values in levels zero through R . Each generator raises the level by one, each adjoint lowers it by one, and an adjoint applied at level zero gives zero. Constant terms preserve level zero. Restricting the evaluated matrix P to the input at level zero therefore gives a fiberwise map

$$P_0 : E_0^{\oplus 2m} \longrightarrow F_{m,R} := \left(\bigoplus_{j=0}^R E_j \right)^{\oplus m} \quad \text{over } Y_{m,R}. \quad (5.3)$$

This is a continuous bundle map. Indeed, each block of the restriction is a finite sum of compositions of the continuous maps $X_i^{(j)}$ and their adjoints, with $j < R$. This description is intrinsic to the bundles; no simultaneous trivialization of the Hilbert fibers or continuity between different tests is required. The continuation beyond level R does not enter these compositions. Figure 1 illustrates this finite propagation and the resulting bundle map.

At each y , let ι_0 include the level-zero input in $\mathcal{H}_{m,R,y}^{\oplus 2m}$, and let ι_F include $F_{m,R}(y)$ in $\mathcal{H}_{m,R,y}^{\oplus m}$. The range statement above says $P(y)\iota_0 = \iota_F P_0(y)$. Evaluation preserves (5.2), so compression of this quadratic-form inequality gives

$$P_0(y)^* P_0(y) = \iota_0^* P(y)^* P(y) \iota_0 \geq \frac{1}{2} I \quad (y \in Y_{m,R}).$$

A word of length at most R , starting at level zero,
stays in these levels

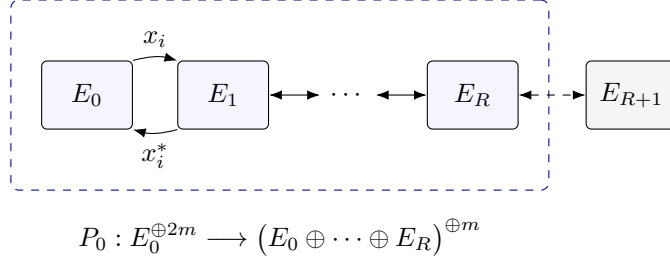


Figure 1: The test indexed by (m, R) , shown schematically. A polynomial matrix P whose entries contain only words of length at most R maps its trivial level-zero input into the displayed target. If $P^*P \geq \frac{1}{2}I$, this restriction would embed a trivial bundle of rank $2md_0$ into that target, contradicting its retained Chern class. The arrows show level changes; each forward block is one of the maps $X_i^{(j)}$. The continuation after level R does not affect this test. When $R = 1$, the labels E_1 and E_R denote the same level.

Hence

$$Q = P_0(P_0^*P_0)^{-1/2}$$

is a continuous fiberwise isometry. Its image is a subbundle of $F_{m,R}$ isomorphic to the trivial bundle $E_0^{\oplus 2m}$, of rank $2m \operatorname{rank}(E_0)$. This contradicts Proposition 4.1. Since m was arbitrary, the conclusion holds for every matrix algebra over B . $\square$

The passage from the two relations to a specified positive pair is useful independently of the construction. It requires only the 1-quasitrace axioms once a quasitrace on the matrix algebra is given.

Lemma 5.2. *Let B be a unital complex C^* -algebra, let $N \geq 1$, and suppose $x_1, \dots, x_N \in B$ satisfy*

$$\sum_{i=1}^N x_i^* x_i = 1, \quad \sum_{i=1}^N x_i x_i^* \leq c1 \quad \text{for some } 0 \leq c < 1.$$

There are positive contractions $a, b \in M_{2N}(B)$, determined by this tuple, such that every normalized 1-quasitrace τ on $M_{2N}(B)$ satisfies

$$\tau(a + b) - \tau(a) - \tau(b) \geq \frac{1 - c}{2N}. \quad (5.4)$$

Proof. Put

$$u_i = x_i^* x_i, \quad v_i = x_i x_i^*, \quad H_i = \sum_{j=1}^i u_j, \quad J_i = \sum_{j=1}^i v_j,$$

with $H_0 = J_0 = 0$. We use these partial sums to define two diagonal matrices. In their additivity defect, the first N blocks will telescope the column partial sums, while the last N use complements to telescope the row partial sums with the opposite sign and keep the entries positive.

$$\begin{aligned} a &= \operatorname{diag}(H_0, \dots, H_{N-1}, 1 - J_0, \dots, 1 - J_{N-1}), \\ b &= \operatorname{diag}(u_1, \dots, u_N, 1 - v_1, \dots, 1 - v_N). \end{aligned} \quad (5.5)$$

All entries are positive contractions: $0 \leq H_i \leq H_N = 1$, $0 \leq u_i \leq 1$, and $0 \leq J_i, v_i \leq J_N \leq c1$.

Fix a normalized 1-quasitrace τ on $M_{2N}(B)$. Write $\iota_k(z)$ for $z \in B$ placed in the k th diagonal corner and define $\varphi(z) = \tau(\iota_1(z))$. If $z \geq 0$, let t be the matrix with $z^{1/2}$ in its (k, ℓ) entry and zeros elsewhere. Then $t^*t = \iota_\ell(z)$ and $tt^* = \iota_k(z)$, so the identity $\tau(t^*t) = \tau(tt^*)$ gives

$$\tau(\iota_k(z)) = \varphi(z) \quad (z \geq 0, 1 \leq k \leq 2N).$$

Since self-adjoint elements supported in distinct diagonal corners commute, abelian linearity yields

$$\tau(\text{diag}(z_1, \dots, z_{2N})) = \sum_{k=1}^{2N} \varphi(z_k) \quad (z_k \geq 0), \quad \varphi(1) = \frac{1}{2N}. \quad (5.6)$$

The last equality follows by taking every $z_k = 1$. The square axiom applied within the first corner also gives

$$\varphi(u_i) = \varphi(v_i) \quad (1 \leq i \leq N). \quad (5.7)$$

The first N diagonal blocks contribute to the additivity defect the terms

$$\varphi(H_i) - \varphi(H_{i-1}) - \varphi(u_i).$$

The last N contribute

$$\begin{aligned} & \varphi(2 - J_i) - \varphi(1 - J_{i-1}) - \varphi(1 - v_i) \\ &= -\varphi(J_i) + \varphi(J_{i-1}) + \varphi(v_i). \end{aligned}$$

Here each scalar-translation identity is taken within the abelian algebra generated by the corner identity and the indicated self-adjoint element. Thus these equalities use no additivity on noncommuting elements of B . Summing the scalar defects, telescoping, and using (5.7), we obtain

$$\begin{aligned} \tau(a + b) - \tau(a) - \tau(b) &= \varphi(H_N) - \sum_{i=1}^N \varphi(u_i) - \varphi(J_N) + \sum_{i=1}^N \varphi(v_i) \\ &= \varphi(1) - \varphi(J_N). \end{aligned}$$

The restriction of φ to $C^*(1, J_N) \subseteq B$ is positive and linear. Since $0 \leq J_N \leq c1$, it follows that $\varphi(J_N) \leq c\varphi(1)$. Equation (5.6) now gives (5.4). $\square$

Proof of Theorem 1.1. Theorem 5.1 supplies the separable unital algebra B , generated by twenty-four elements satisfying (5.1), with no properly infinite matrix algebra over it. Consequently the same is true of $A = M_{48}(B)$, because $M_k(A) \cong M_{48k}(B)$ for every k . Theorem 2.1 gives a normalized 2-quasitrace on A .

Apply Lemma 5.2 with $N = 24$ and $c = 2/3$. It supplies positive contractions $a, b \in A$ for which every normalized 2-quasitrace has defect at least $(1 - c)/(2N) = 1/144$. Matrix amplification preserves separability and unitality, so all assertions of Theorem 1.1 follow. $\square$

6 Tensor products and stable finiteness

The consequences in Corollary 1.2 use the established factor form of the quasitrace problem and a trace-obstruction theorem for tensor products. We record their scopes before applying them to the different algebras in the corollary.

The factor input. The assertion that every normalized 2-quasitrace on every unital C^* -algebra is a tracial state is equivalent to the assertion that every AW^* -factor of type II_1 is a W^* -algebra; see [8, Section 2] and [4, Theorem 4.11]. Every such factor admits a normalized 2-quasitrace, and it admits a tracial state exactly when it is a W^* -algebra [8, Section 2]. A finite AW^* -factor is simple as a C^* -algebra [4, Remark 2.7]. For a simple unital C^* -algebra, absence of stable proper infiniteness is equivalent to stable finiteness [8, Section 2]. These statements impose no separability condition on the factor.

The tensor input and the passage to $\mathbb{F}_2$. Haagerup's trace-obstruction theorem, in the properly infinite formulation recorded by Milhøj and Rørdam [8, Theorem 3.1], says that a unital C^* -algebra D has no tracial state if and only if $D \otimes_{\min} C_r^*(\mathbb{F}_\infty)$ is properly infinite. Haagerup notes that the subgroup inclusion $\mathbb{F}_\infty \hookrightarrow \mathbb{F}_2$ induces a unital inclusion $C_r^*(\mathbb{F}_\infty) \hookrightarrow C_r^*(\mathbb{F}_2)$ [6, Remark 2.5(a)]. The minimal tensor product preserves injective $*$ -homomorphisms, so it gives a unital inclusion

$$D \otimes_{\min} C_r^*(\mathbb{F}_\infty) \hookrightarrow D \otimes_{\min} C_r^*(\mathbb{F}_2).$$

The two isometries witnessing proper infiniteness retain their relations under this inclusion. Consequently,

$$D \text{ has no tracial state} \implies D \otimes_{\min} C_r^*(\mathbb{F}_2) \text{ is properly infinite.} \quad (6.1)$$

Proof of Corollary 1.2. For (i), Theorem 1.1 and the factor input give an AW^* -factor M of type II_1 that is not a W^* -algebra. Hence M has no tracial state. It is simple and admits a normalized 2-quasitrace. Theorem 2.1 therefore excludes stable proper infiniteness, and the criterion for simple algebras makes M stably finite.

Powers's theorem [9] gives simplicity of $R = C_r^*(\mathbb{F}_2)$. Its canonical tracial state is faithful, as are the matrix traces induced by it. A faithful trace rules out a nonunitary isometry, so R is stably finite. It is also unital. Applying (6.1) to M proves (i): $M \otimes_{\min} R$ is properly infinite.

For (ii), take the explicit algebra B of Theorem 5.1. No matrix algebra over B is properly infinite, so Theorem 2.1 supplies a normalized 2-quasitrace. The relations (5.1) exclude a tracial state: applying one to their two sums would give $1 \leq 2/3$. Equation (6.1) makes $B \otimes_{\min} R$ properly infinite, and Theorem 2.1 then excludes a normalized 2-quasitrace on that tensor product.

For (iii), Haagerup's finite trace-obstruction criterion [6, Lemma 2.1] supplies $y_1, \dots, y_n \in M$ such that

$$\sum_{i=1}^n y_i^* y_i = 1, \quad \left\| \sum_{i=1}^n y_i y_i^* \right\| < 1.$$

Let $C = C^*(1, y_1, \dots, y_n) \subset M$. This algebra is separable and unital. It inherits stable finiteness from M : an isometry in $M_k(C)$ is an isometry in $M_k(M)$, hence is unitary there and therefore already unitary in $M_k(C)$. A tracial state τ on C would satisfy

$$1 = \tau \left(\sum_{i=1}^n y_i^* y_i \right) = \tau \left(\sum_{i=1}^n y_i y_i^* \right) \leq \left\| \sum_{i=1}^n y_i y_i^* \right\| < 1,$$

a contradiction. This proves (iii). $\square$

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